

Representation Invariance and Structural Minimality of a Six-Component Universal Kernel

Abstract

We study a six-component kernel consisting of two unary and three binary relational structures together with two quantitative binary observables. We formalize representation invariance under bijections of the underlying carrier and prove that the resulting transport operation is compatible with composition, identity, and inverse transport. We further establish structural minimality for the six primitive components and prove invariance of finite relational paths, finite reachability, and the induced operations on state sets.

All statements are purely structural. No probabilistic axiom, Born rule, Hilbert-space structure, Gleason theorem, or numerical constant is assumed.

1 The universal kernel

Let X be a type of states. A universal kernel on X consists of six primitive components

$$U = (M_U, G_U, A_U, \Psi_U, \Lambda_U, \text{Proj}_U),$$

where

$$M_U : X \rightarrow \text{Prop},$$

$$G_U, A_U, \text{Proj}_U : X \times X \rightarrow \text{Prop},$$

and

$$\Psi_U, \Lambda_U : X \times X \rightarrow \mathbb{R}.$$

The construction is intentionally representation-independent: the elements of X carry no canonical names.

Definition 1.1 (Universal kernel). A *universal kernel* on X is a tuple

$$U = (M_U, G_U, A_U, \Psi_U, \Lambda_U, \text{Proj}_U)$$

with the types specified above.

2 Representation equivalence

Suppose that U is a kernel on X and V is a kernel on Y .

Definition 2.1 (Representation equivalence). We say that U and V are *representation equivalent*, written

$$U \sim_{\text{rep}} V,$$

if there exists a bijection

$$e : X \xrightarrow{\sim} Y$$

such that, for every $x, y \in X$,

$$\begin{aligned} M_U(x) &\iff M_V(e(x)), \\ G_U(x, y) &\iff G_V(e(x), e(y)), \\ A_U(x, y) &\iff A_V(e(x), e(y)), \\ \Psi_U(x, y) &= \Psi_V(e(x), e(y)), \\ \Lambda_U(x, y) &= \Lambda_V(e(x), e(y)), \end{aligned}$$

and

$$\text{Proj}_U(x, y) \iff \text{Proj}_V(e(x), e(y)).$$

The crucial point is that equivalence concerns the complete representation, rather than the equality of the underlying carrier types.

Theorem 2.2 (Reflexivity). *For every kernel U ,*

$$U \sim_{\text{rep}} U.$$

Proof. Take the identity bijection of the carrier. Every component is then preserved pointwise. $\square$

Theorem 2.3 (Symmetry). *If*

$$U \sim_{\text{rep}} V,$$

then

$$V \sim_{\text{rep}} U.$$

Proof. Let $e : X \rightarrow Y$ be a witnessing bijection. Its inverse $e^{-1} : Y \rightarrow X$ witnesses the reverse equivalence. Each of the six preservation conditions is reversed using the symmetry of logical equivalence and equality. $\square$

Theorem 2.4 (Transitivity). *If*

$$U \sim_{\text{rep}} V \quad \text{and} \quad V \sim_{\text{rep}} W,$$

then

$$U \sim_{\text{rep}} W.$$

Proof. Let

$$e : X \xrightarrow{\sim} Y, \quad f : Y \xrightarrow{\sim} Z$$

witness the two equivalences. The composite

$$f \circ e : X \xrightarrow{\sim} Z$$

witnesses the desired equivalence. For example,

$$G_U(x, y) \iff G_V(e(x), e(y)) \iff G_W(f(e(x)), f(e(y))),$$

and the other five components are identical in form. $\square$

3 Canonical transport

Given a bijection

$$e : X \xrightarrow{\sim} Y$$

and a kernel U on X , define its transport to Y by

$$e_*U = (M', G', A', \Psi', \Lambda', \text{Proj}'),$$

where

$$\begin{aligned} M'(y) &\iff M_U(e^{-1}(y)), \\ G'(y_1, y_2) &\iff G_U(e^{-1}(y_1), e^{-1}(y_2)), \end{aligned}$$

and similarly

$$\begin{aligned} A'(y_1, y_2) &\iff A_U(e^{-1}(y_1), e^{-1}(y_2)), \\ \Psi'(y_1, y_2) &= \Psi_U(e^{-1}(y_1), e^{-1}(y_2)), \\ \Lambda'(y_1, y_2) &= \Lambda_U(e^{-1}(y_1), e^{-1}(y_2)), \\ \text{Proj}'(y_1, y_2) &\iff \text{Proj}_U(e^{-1}(y_1), e^{-1}(y_2)). \end{aligned}$$

Theorem 3.1 (Canonical representation invariance). *For every kernel U on X and every bijection*

$$e : X \xrightarrow{\sim} Y,$$

one has

$$U \sim_{\text{rep}} e_*U.$$

Proof. The witnessing bijection is precisely e . Each preservation condition follows immediately from the definition of transport and the identities

$$e^{-1}(e(x)) = x.$$

□

4 Functoriality of transport

Transport behaves as expected with respect to identities and composition.

Theorem 4.1 (Identity transport). *For every kernel U ,*

$$(\text{id}_X)_*U = U.$$

Theorem 4.2 (Composition of transports). *Let*

$$e : X \xrightarrow{\sim} Y, \quad f : Y \xrightarrow{\sim} Z.$$

Then

$$f_*(e_*U) = (f \circ e)_*U.$$

Proof. Both kernels agree pointwise on all six primitive components. For example,

$$\begin{aligned} G_{f_*(e_*U)}(z_1, z_2) &\iff G_{e_*U}(f^{-1}z_1, f^{-1}z_2) \\ &\iff G_U(e^{-1}f^{-1}z_1, e^{-1}f^{-1}z_2) \\ &\iff G_U((f \circ e)^{-1}z_1, (f \circ e)^{-1}z_2). \end{aligned}$$

The remaining components are identical.

□

Theorem 4.3 (Inverse transport). *For every bijection*

$$e : X \xrightarrow{\sim} Y,$$

$$e_*^{-1}(e_*U) = U.$$

Proof. Apply the previous theorem to e and e^{-1} and use

$$e^{-1} \circ e = \text{id}_X.$$

□

5 Structural minimality

We now distinguish representation invariance from minimality. The latter concerns whether a primitive component can be removed without changing the kernel.

Let

$$U^{\setminus M}, \quad U^{\setminus G}, \quad U^{\setminus A}, \quad U^{\setminus \Psi}, \quad U^{\setminus \Lambda}, \quad U^{\setminus \text{Proj}}$$

denote the corresponding modified kernels.

Theorem 5.1 (Structural necessity). *For the designated test kernel U_0 ,*

$$U_0^{\setminus M} \neq U_0,$$

$$U_0^{\setminus G} \neq U_0,$$

$$U_0^{\setminus A} \neq U_0,$$

$$U_0^{\setminus \Psi} \neq U_0,$$

$$U_0^{\setminus \Lambda} \neq U_0,$$

and

$$U_0^{\setminus \text{Proj}} \neq U_0.$$

Proof. Each inequality is witnessed by evaluating the modified and original kernels at a suitable state or pair of states. The corresponding component takes different values, and hence the kernels cannot be equal. □

Corollary 5.2 (Six-component minimality). *None of the six designated primitive components can be deleted from the test kernel while preserving the resulting kernel structure.*

6 Finite relational paths

Let R be a binary relation on X . Define the finite path relation recursively by

$$\text{Path}_0^R(x, z) \iff x = z,$$

and

$$\text{Path}_{n+1}^R(x, z) \iff \exists y \in X, R(x, y) \wedge \text{Path}_n^R(y, z).$$

Theorem 6.1 (Invariance of finite paths). *Suppose*

$$U \sim_{\text{rep}} V$$

via $e : X \rightarrow Y$. *Then for every* $n \in \mathbb{N}$,

$$\text{Path}_n^{G_U}(x, z) \iff \text{Path}_n^{G_V}(e(x), e(z)).$$

Proof. By induction on n . The case $n = 0$ follows from

$$e(x) = e(z) \iff x = z.$$

For the successor case, transport the intermediate state y to $e(y)$ and use the induction hypothesis. The converse uses e^{-1} . $\square$

Theorem 6.2 (Invariance of finite A -paths). *Under the same hypotheses,*

$$\text{Path}_n^{A_U}(x, z) \iff \text{Path}_n^{A_V}(e(x), e(z)).$$

Theorem 6.3 (Invariance of finite Proj-paths). *Under the same hypotheses,*

$$\text{Path}_n^{\text{Proj}_U}(x, z) \iff \text{Path}_n^{\text{Proj}_V}(e(x), e(z)).$$

7 Finite reachability

Define

$$\text{Reach}_{\text{Proj}_U}(x, z) \iff \exists n \in \mathbb{N}, \text{Path}_n^{\text{Proj}_U}(x, z).$$

Theorem 7.1 (Representation invariance of reachability). *If*

$$U \sim_{\text{rep}} V$$

via e , *then*

$$\text{Reach}_{\text{Proj}_U}(x, z) \iff \text{Reach}_{\text{Proj}_V}(e(x), e(z)).$$

Proof. Choose the same path length n on both sides and apply the invariance of finite Proj-paths. $\square$

8 Transport of state sets

For $S \subseteq X$, define

$$e_*S = e[S].$$

Theorem 8.1 (Preservation of membership). *For every* $S \subseteq X$,

$$x \in S \iff e(x) \in e_*S.$$

Proof. The forward implication follows from the witness x in the definition of the image. Conversely, if $e(x) \in e[S]$, there exists $y \in S$ such that

$$e(y) = e(x).$$

Injectivity of e gives $y = x$, hence $x \in S$. $\square$

Theorem 8.2 (Preservation of unions). *For all $S, T \subseteq X$,*

$$e_*(S \cup T) = e_*S \cup e_*T.$$

Theorem 8.3 (Preservation of intersections). *For all $S, T \subseteq X$,*

$$e_*(S \cap T) = e_*S \cap e_*T.$$

Proof. The union statement follows directly from the definition of image. For intersections, if a point belongs to both images, injectivity of e identifies the two corresponding preimages, which therefore belong simultaneously to S and T . $\square$

9 Formal verification

The preceding results admit direct formalization in Lean. The formal development uses a propositional relation

$$\text{UFRepresentationEquivalent}(U, V)$$

rather than identifying it with the structure ‘UFEquiv’, since the latter is a record carrying an equivalence together with its preservation fields.

The formal verification establishes:

- (i) reflexivity, symmetry, and transitivity of representation equivalence;
- (ii) canonical invariance under carrier transport;
- (iii) identity, composition, and inverse laws for transport;
- (iv) structural necessity of all six primitive components;
- (v) invariance of finite G , A , and Proj paths;
- (vi) invariance of finite Proj reachability;
- (vii) invariance of the induced state-set operations.

No probabilistic or Hilbert-space structure enters any of these arguments.

10 Conclusion

The formal results establish a precise structural notion of representation invariance for the six-component kernel.

The central conclusion is that a change of representation of the carrier does not alter the kernel’s primitive content. Moreover, this invariance propagates to finite relational constructions and to the induced algebra of state sets.

At the same time, the minimality theorem shows that, for the specified test kernel, each of the six primitive components contributes genuinely distinct structural information.

These conclusions are entirely independent of probability theory, the Born rule, Hilbert-space geometry, Gleason-type reconstruction, or any distinguished numerical constant.

A Lean verification

The complete machine-checkable proofs corresponding to the statements above are supplied separately as a Lean source file. The mathematical manuscript and the formal verification should be regarded as complementary: the former presents the conceptual structure, while the latter certifies the individual propositions in a proof assistant.